

HACKATHON SEASON 3 – MISSION 7

TASK: 1. Identify one inefficient process (2+ hours weekly)

2. Create AI solution prototype using existing skills

3. Document time/cost savings with metrics

4. Schedule meeting with manager

AI-Powered HR or IS Support System: Cost/Time-Benefit Analysis Report

Transforming Employee Service Delivery Through Intelligent Automation

Objective

This document compares the traditional HR/IS service request process with an AI-powered support system built using **Chatling AI** and **Make.com**. It highlights measurable **time and cost savings**, improved response efficiency, and overall productivity enhancements across HR service functions.

Section 1: Executive Summary

Traditional corporate HR/IS service handling involves manual workflows, repetitive follow-ups, and delayed responses, leading to substantial productivity loss. Employees often spend hours waiting for clarity or approvals on routine queries related to onboarding, insurance, transfers, and leaves.

To solve this, an **AI-powered system** was implemented with:

- A **Chatling AI chatbot** to instantly answer FAQs and collect structured requests
- **Make.com automation** to route submissions, store logs, trigger reminders after 3 hours, and escalate to higher authorities if unresolved after 5 hours

Core benefits:

-  **Significantly faster resolution**
-  **Reduced HR operational cost**
-  **Improved tracking and SLA compliance**

-  Less manual follow-up by employees

Section 2: Traditional vs AI-Driven Support – Comparison Table

Factor	Traditional Process	AI-Powered System (Chatbot + Make.com)
Avg. Response Time (per query)	4–12 hours	0.5–1 hour
Manual follow-ups per request	2–3	0 (automated alerts + escalation)
Support Staff Required	Full-time HR/IS agents	30–50% reduction in load
Employee Time Lost/Request	1–2 hrs	5–10 mins
HR Ops Time Spent/Request	10–20 mins	2–3 mins
Error Rate in Handling	Medium–High (manual entry)	Low (automated + structured logging)
Monthly Operational Cost	₹1,00,000+ (for 2 agents)	₹8,000–₹12,000 (Chatling + Make.com)

Assumptions:

AvgHR Salary = ₹50K/month

Employee cost of delay = ₹200/hr

Tools cost = Chatling (₹3,000), Make.com (₹5,000–₹8,000) depending on volume

Section 3: System Overview

The AI-powered system consists of:

Chatling AI Chatbot

- Answers employee queries 24/7
- Supports services like Onboarding, Transfers, Leaves, Health Checkups, etc.
- Gathers form inputs for requests

Make.com Automation

- Captures form data via webhook
- Routes to HR/IS on Discord (based on request type)
- Logs data to Google Sheets (for timestamp tracking)
- Sends automated **reminder after 3 hours**
- Sends **escalation to senior HR/IS** after 2 days if still unresolved

 Section 4: Time Efficiency Insights

Scenario	Without AI	With AI System
100 service requests/month	~150 hours wasted	~10–15 hours total
Avg. resolution delay	4–12 hours	< 1 hour
Follow-up calls/emails avoided	200+	100% automated

-  **Estimated Time Saved/Month:** ~130–140 hours
-  **Employee productivity gain:** Better focus, less frustration
-  **Faster SLA compliance** → smoother HR operations

 Section 5: Cost Savings Estimation

 Monthly Cost Comparison

Element	Traditional (Manual)	AI System
HR Ops (2 FTEs @ ₹50k)	₹1,00,000	₹0 (load shifted)
Employee Time Lost (₹200/hr)	₹30,000+	< ₹3,000
Tools (Chatling + Make)	₹0	₹8,000–₹12,000
Total Monthly Cost	₹1,30,000+	₹11,000–₹15,000

 Net Savings:

-  ₹1.2L+ saved monthly
 -  90% efficiency boost
-

□ Section 6: Industry Comparison

Metric	Industry Average AI System Performance	
Response Time (HR queries)	4–8 hours	0.5–1 hour
HR to Employee Ratio	1:80–1:120	1:150+ (due to AI load handling)
SLA Breach Rate	10–25%	< 2%
Escalation Handling Time	2–5 days	Automated in < 5 hrs

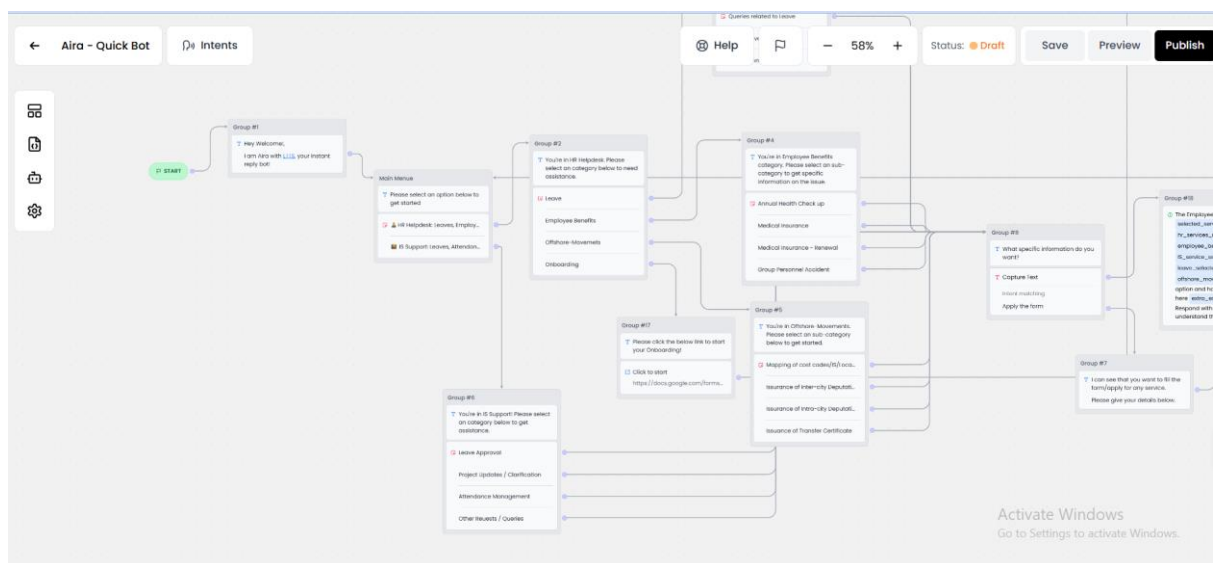
Top companies (Google, Amazon, Deloitte) use AI chatbots + workflow automation to improve HR helpdesk efficiency. Your solution matches their **best practices at a fraction of the cost.**

1. CHAT BOT:

Chatbot Name: Aira – Quick Bot

Tool: Chatling AI

Builder:



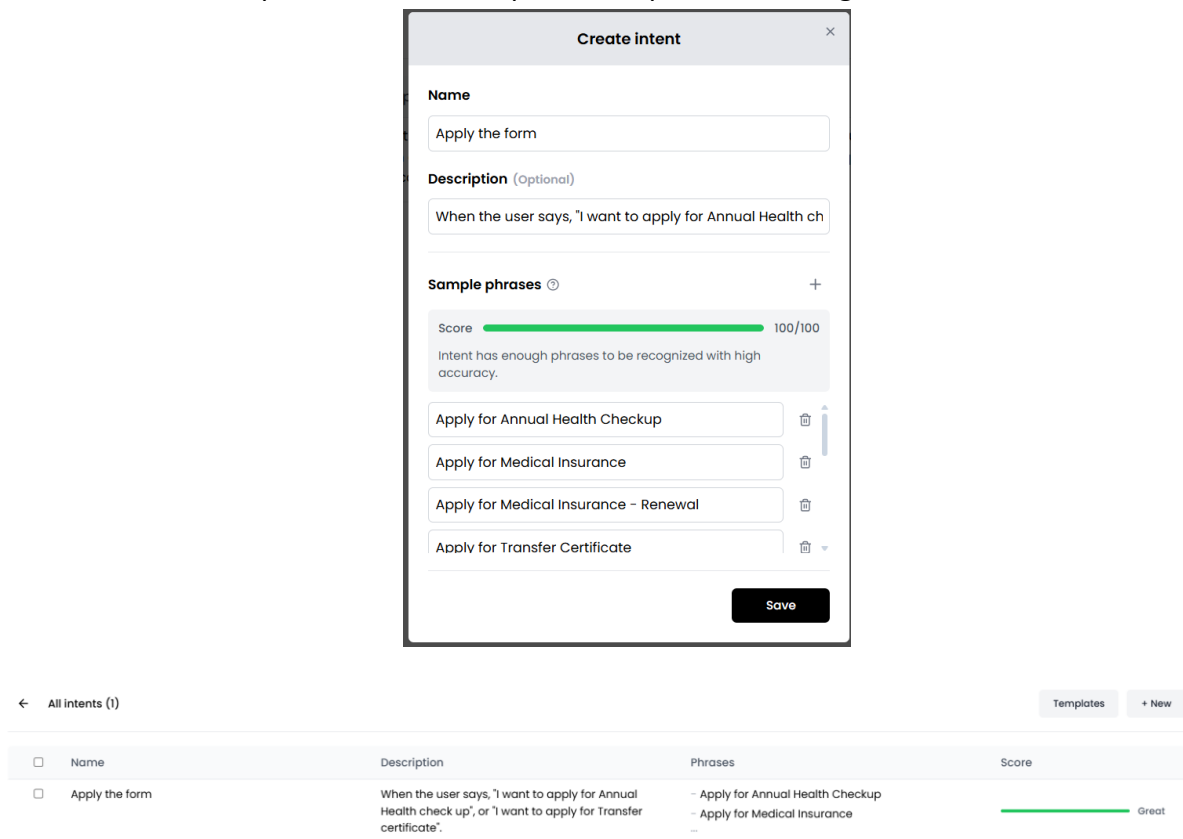
Services:

- **HR Helpdesk**
 - ❖ Leave – Queries related to leave, Gift a Leave, PL Encashment
 - ❖ Employee Benefits – Annual Health check-up, Medical Insurance, Medical Insurance-Renewal, Group Personnel Accident

- ❖ Offshore Movements – Mapping of cost codes/IS/Location, Issuance of Inter-city deputation Letter, Issuance of Intra-city deputation Letter, Issuance of Transfer Certificate
- ❖ Onboarding – Onboarding of New Employees
- **IS Helpdesk**
 - ❖ Leave Approval
 - ❖ Project Updates/Clarifications
 - ❖ Attendance Management
 - ❖ Other Requests/Queries

Features Used:

- **Intents** - Intents identify what users are trying to accomplish and capture the meaning behind their messages, regardless of how they phrase them.
- For example, when users ask, “I want to apply for leave” or “Apply for Annual health check-up”, they share the same underlying goal - to show them form to fill the details.
- Intents can be used to define how your chatbot should respond when it recognizes these user goals. Each intent can be linked to specific conversation flows, data collection requirements, and response templates, enabling the chatbot.



The image shows a 'Create intent' dialog box and a list of created intents.

Create intent dialog box:

- Name:** Apply the form
- Description (Optional):** When the user says, "I want to apply for Annual Health ch
- Sample phrases:**
 - Score: 100/100 (Green bar)
 - Intent has enough phrases to be recognized with high accuracy.
 - Apply for Annual Health Checkup
 - Apply for Medical Insurance
 - Apply for Medical Insurance – Renewal
 - Apply for Transfer Certificate
- Save** button

All intents (1) table:

Name	Description	Phrases	Score
Apply the form	When the user says, "I want to apply for Annual Health check up", or "I want to apply for Transfer certificate".	- Apply for Annual Health Checkup - Apply for Medical Insurance	Great (Green bar)

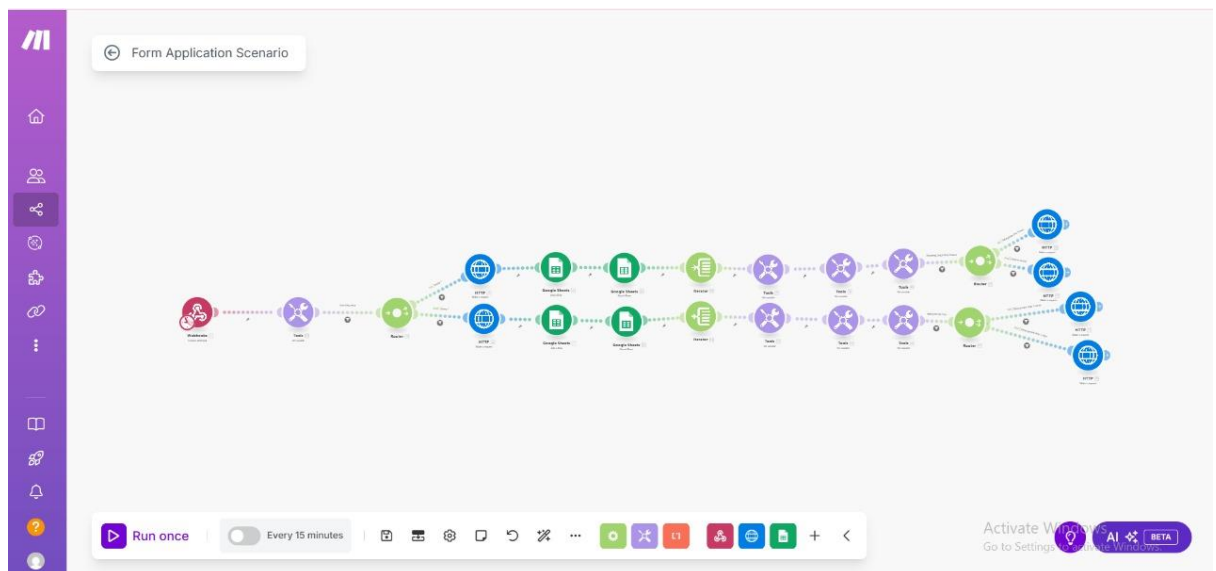
Link for Aira – Quick bot: <https://share.chatling.ai/s/nLyUX2zr147JXb3>

Password: quickbot@ai

2. AUTOMATION

Tool: Make.com

Scenario 1: Application Form



Scenario 1 gathers the details filled by employee via webhook.

- Routes the messages to HR Team and IS Team based on the services selected by the employee.
- I have created 2 discord channels, one for HR Team and another for IS Team. Anyone who is available in both teams should respond to the queries requested by employees.
- When employees fill the form in chatbot, the query message is delivered to HR / IS teams immediately.
- Details are also stored in google sheets for reference and extra column "Response Status" for tracking the time and response status.
- I have tracked the time difference between request time and response time using Tools module's Set Variable action.
- If response is not received after 3 hours of request time, a "Remainder Alert" is sent to the respective discord channel.
- If response is not received even after 2days of request time, "Escalation message" is sent to higher authority.

When service is HR Service, the below path of scenario will run:



When the service is IS Service, the below path will run:



Modules Configuration and more details:

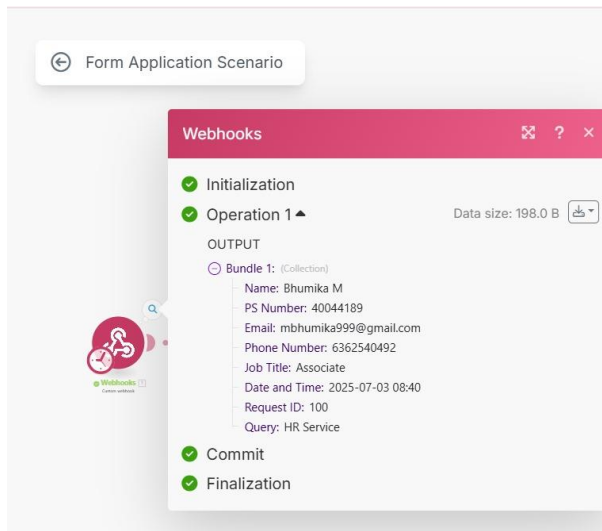


Fig. 1

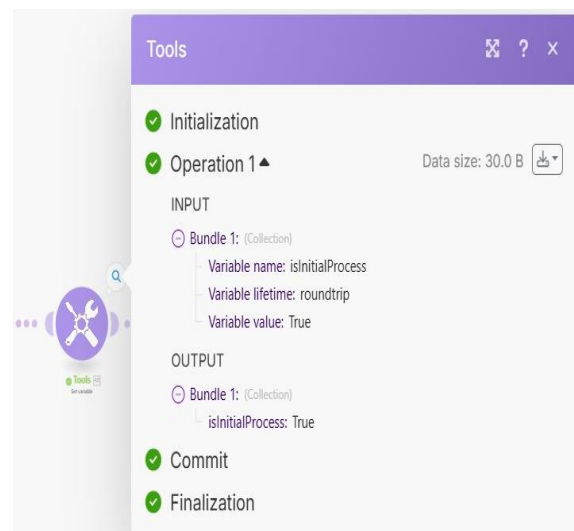


Fig. 2

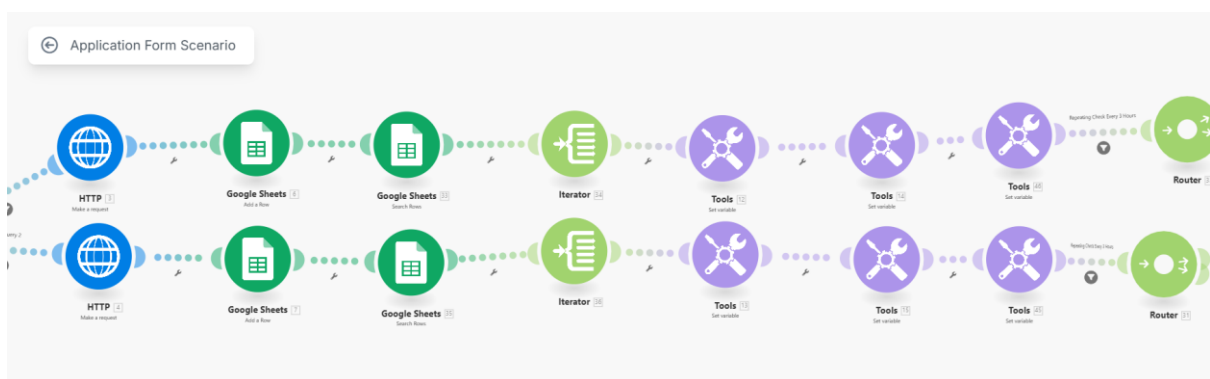


Fig. 3

Webhook (gathers details filled by employees) → Tools (Set a Variable called IsInitialProcess = True) → Router (To transfer the data to two different paths based on service as HR Service or IS Service)

HR Service / IS Service:

If IsInitialProcess = True, the below process runs,

Router → HTTP (Post request, To send message to HR Team discord channel) → Google Sheets (Add a Row, To add all the details to google sheets for tracking purpose) → Google sheets (Search a Row, Searches a row based on “Response Status = Pending”) → Iterator (Iterates throw each row given my search a row module) → Tools(Set a Variable, To calculate the delay time using “now” variable and formatted it to YYYY-MM-DD HH:mm format) → Tools (Set a Variable, To find the time difference between request time and delay time) → Tools (Set a Variable, Resetting is InitialProcess = False) → Router (To routing the data based on delay response).

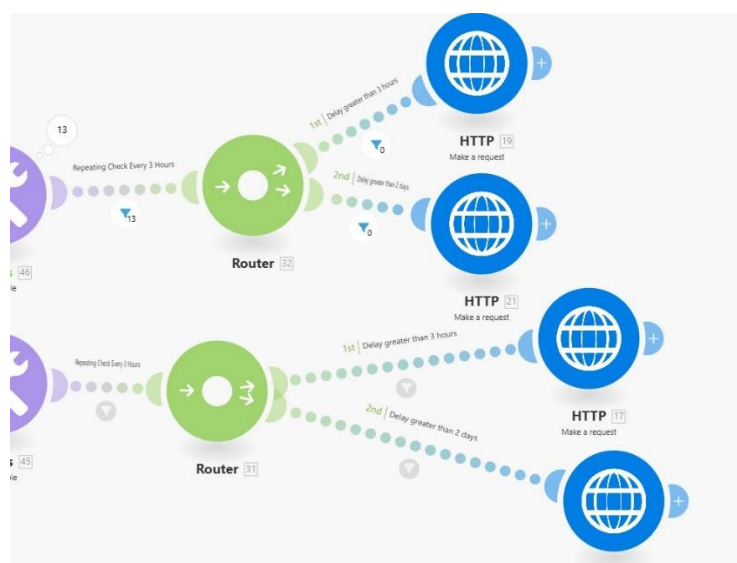


Fig. 4

If response time is more than 3 hours, Remainder alert is sent, if response time exceeds 2 days, Escalation alert is sent to higher authority (Ref. Fig. 4).

Scenario 2: Response Time and Approval

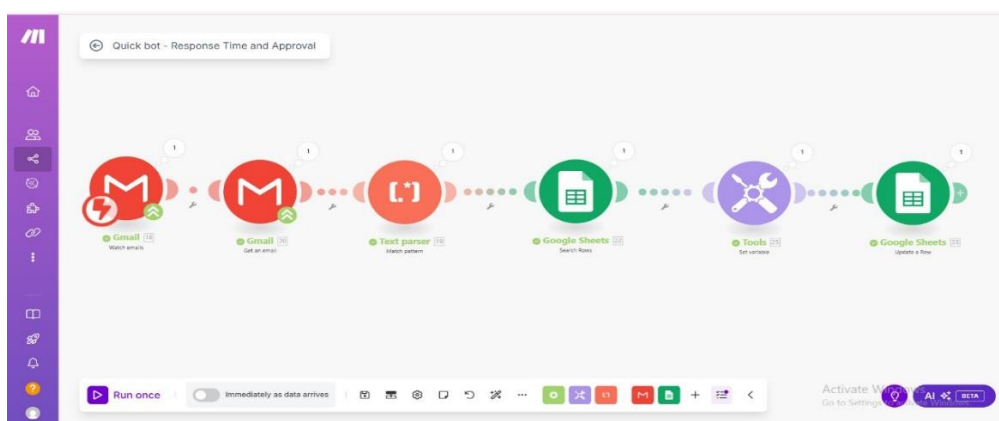


Fig. 5

Modules Configuration:

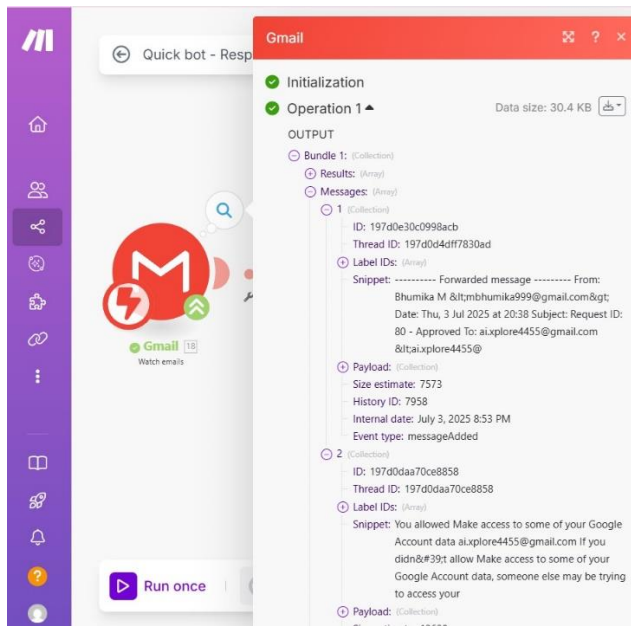


Fig. 6

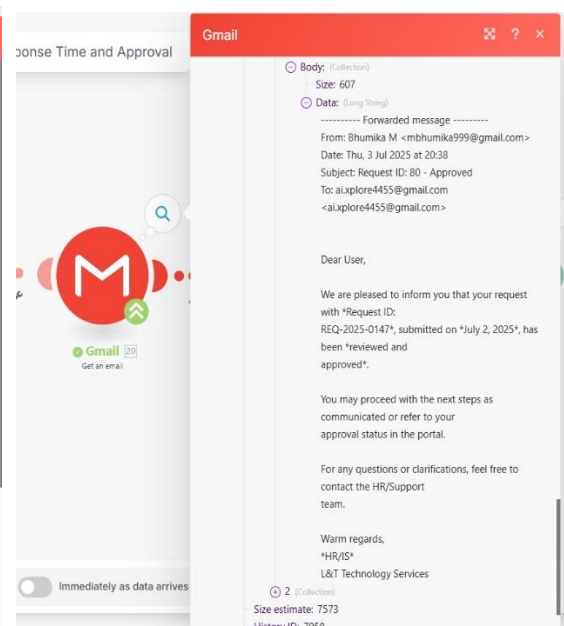


Fig. 7

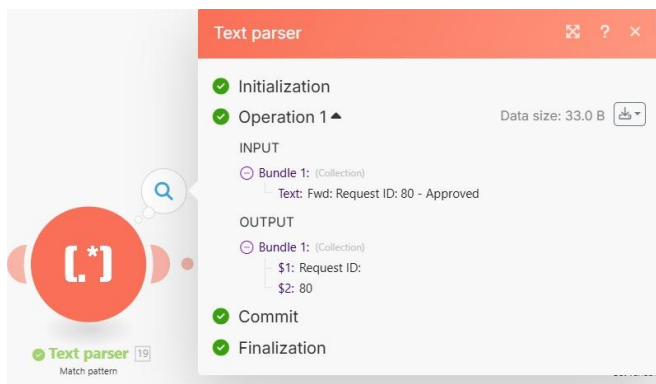


Fig. 8

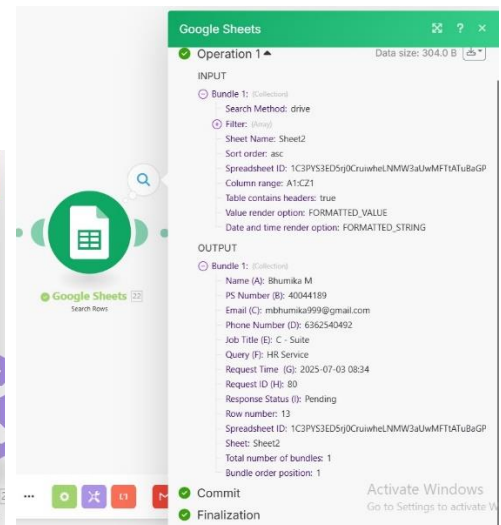


Fig. 9

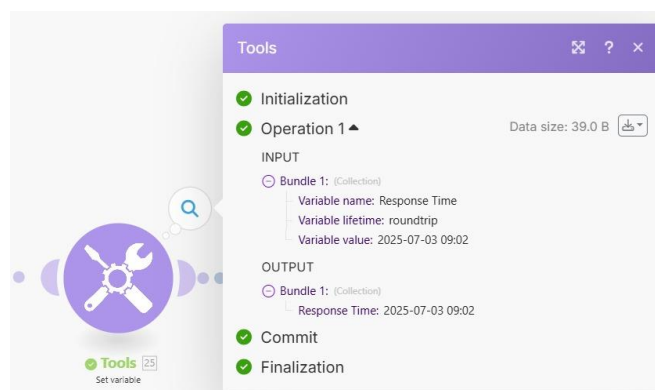


Fig. 10

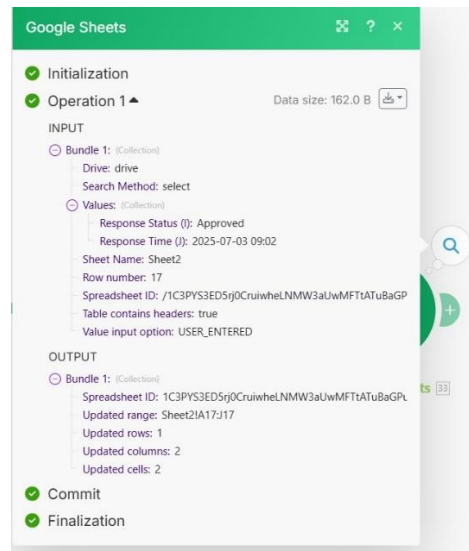


Fig. 11

Scenario 3: Onboarding Assistant

- Google Form (Watch Response) – Google form is created to receive the details from the employee
- Tools (Set a Variable) – As discord channels will only accept the name as lower letters. The channel name (name of the employee) is converted to lower letters using lower ().
- Discord (Create a Guild channel) – Created a discord channel to each employee
- Discord (Create a channel invite) – Created a channel invite to invite employees to the discord channel.
- Gmail (Send an email) – Welcome Email is sent to each employee.
- HTTP (Make a POST request, A voice bot is triggered and call is made to the employee to handle further process of onboarding)

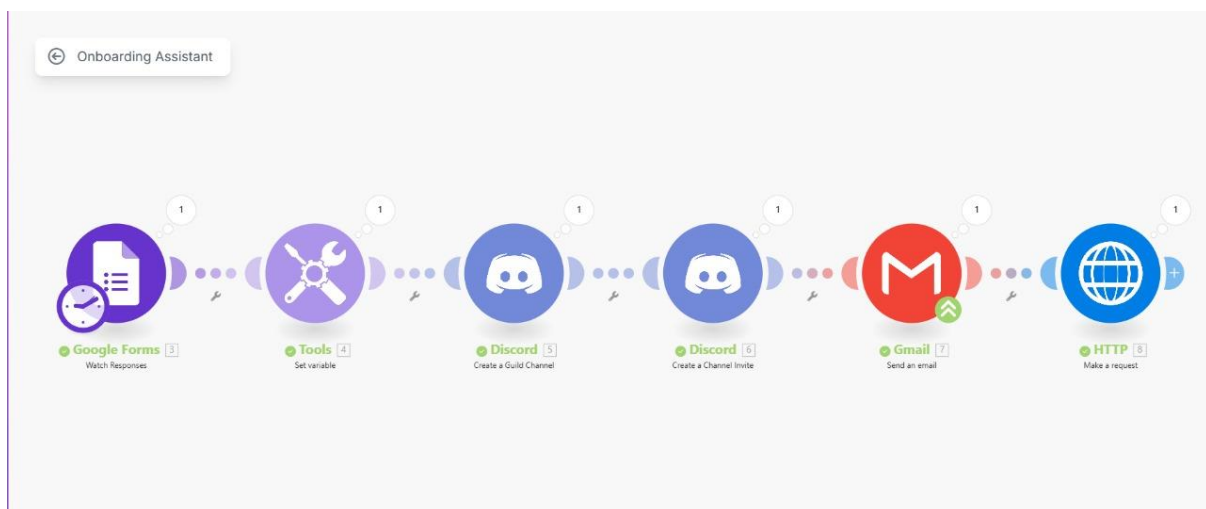


Fig. 12

Module Configuration:

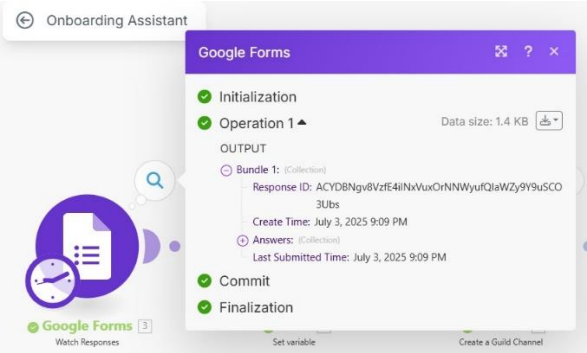


Fig. 13

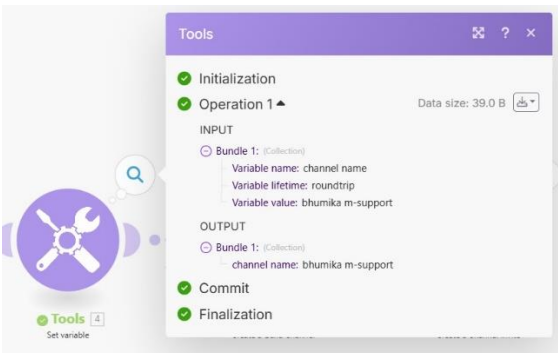


Fig. 14

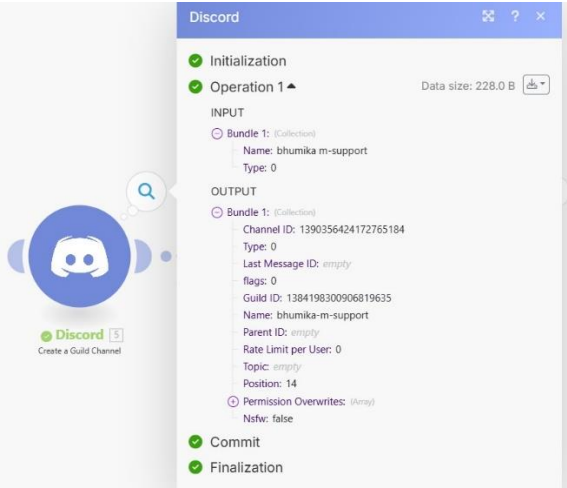


Fig. 15

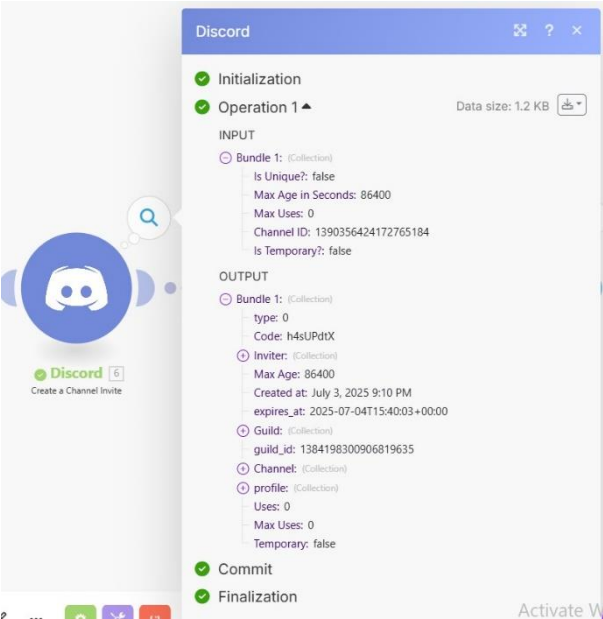


Fig. 16

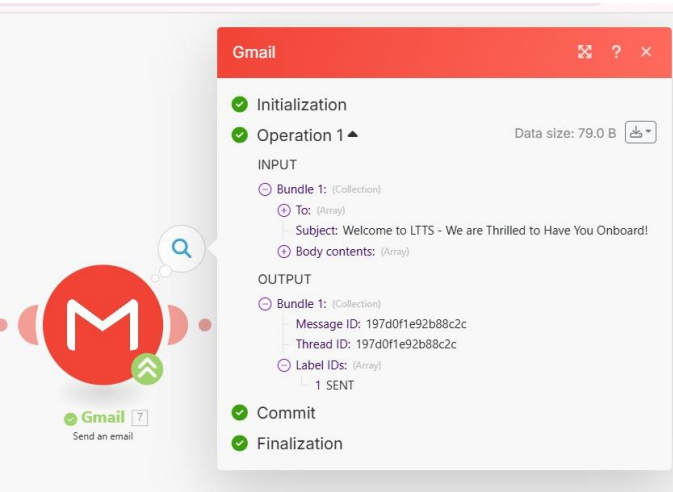


Fig. 17

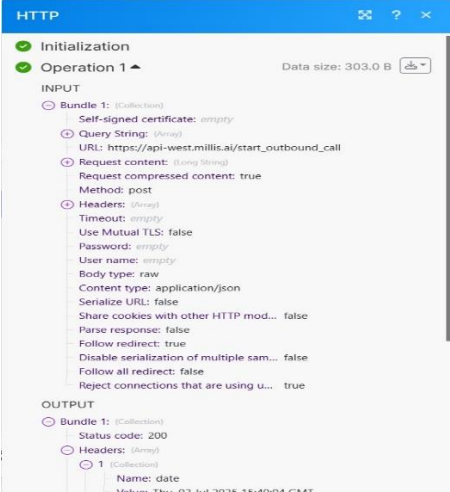


Fig. 18

Voice Bot – Ben

Tool: Milis AI

Ben – Onboarding Assistant:

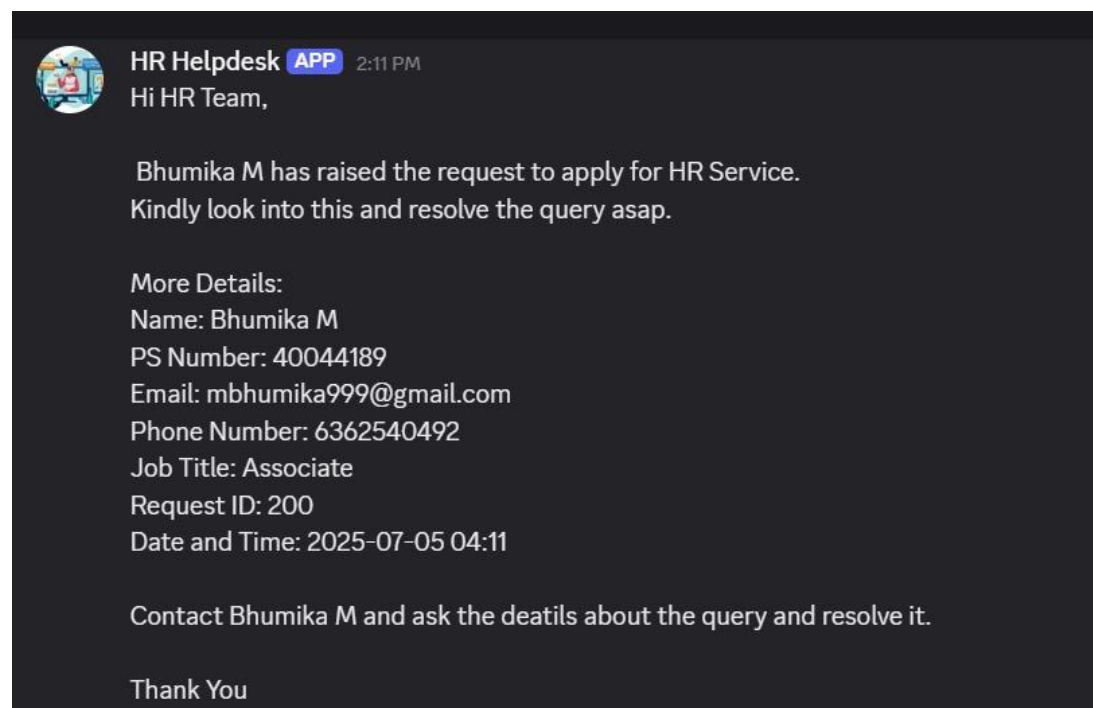
Ben does similar work to Alex, but its main responsibility is to handle onboarding of new employees. Ben collects basic information from employees (Name, Email, contact no., Job Title) by assisting them to fill in the form and immediately triggers the outbound call to each one of the employees to book call with the Manager, to assist them with all the basic information about the company and even answering all the queries relating to the company.

Link for Ben – Onboarding Assistant:

<https://app.millis.ai/agents/embedded?id=OSVFYkZOY3M04SYqBf&k=YBFGuHxSsowkAROCKBwlBr4ijE4ZBC8l>

Results:

- **Response Delay**
- ❖ **A Request Message delivered to HR Helpdesk.**



❖ A Request message delivered to IS Helpdesk

**IS Helpdesk** APP 7/3/25, 8:28 PM
Hey IS Team,

Bhumika M has raised the request for IS Service. Kindly looking into it and resolve the issue asap.

More Details:
Name: Bhumika M
PS Number: 40044189
Email: mbhumika999@gmail.com
Phone Number: 6362540492
Job Title: Associate
Request ID: 70
Date and Time: 2025-07-03 08:28

Contact Bhumika M and gather more details and resolve the issue.

Thank you

❖ Remainder alert is sent to the HR / IS team based on the service if there is no response after 3 hours of request time.

Please take necessary action or respond at your earliest convenience.

 Auto-Reminder: Request ID: 4001, Query: HR Service submitted at 2025-07-03 06:00 is still pending.

Please take necessary action or respond at your earliest convenience.

❖ CHRO Team is created in discord to handle Escalation Emails and messages

chro-team     Search



Welcome to #chro-team!

This is the start of the #chro-team channel.

 [Edit Channel](#)

 | Message #chro-team

❖ Google Sheet Data:

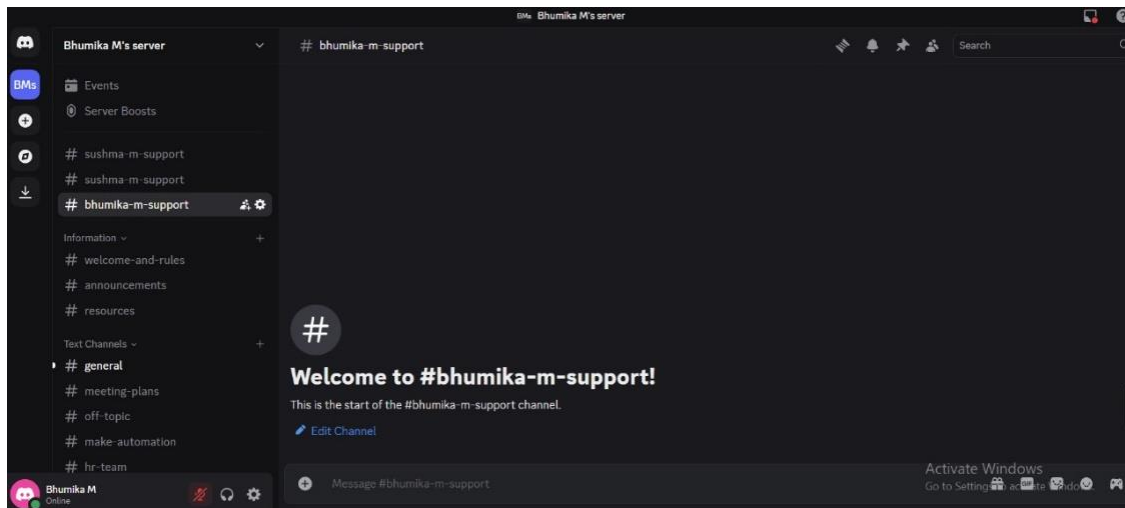
QuickReply Assistant

File Edit View Insert Format Data Tools Extensions Help

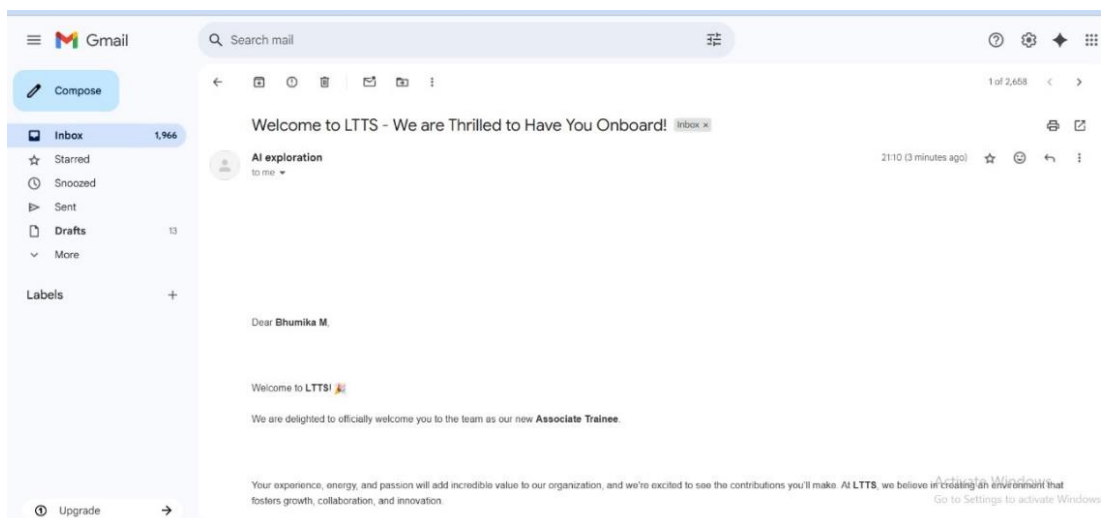
Close deals faster with eSignatures. Send and collect digital signatures directly from PDFs and Docs – plus get 2 TB of storage and more. [Try Workspace](#)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Name	PS Number	Email	Phone Number	Job Title	Query	Request Time	Request ID	Response Status	Response Time				
2	Bhumika M	40044189	mbhumika999@	6362540492	Associate	HR Service	2025-07-05 04:11	200	Approved	2025-07-05 02:48				
3	Bhumika M	40044189	mbhumika999@	6362540492	Associate	IS Service	2025-07-05 02:46	204	Approved	2025-07-05 02:50				
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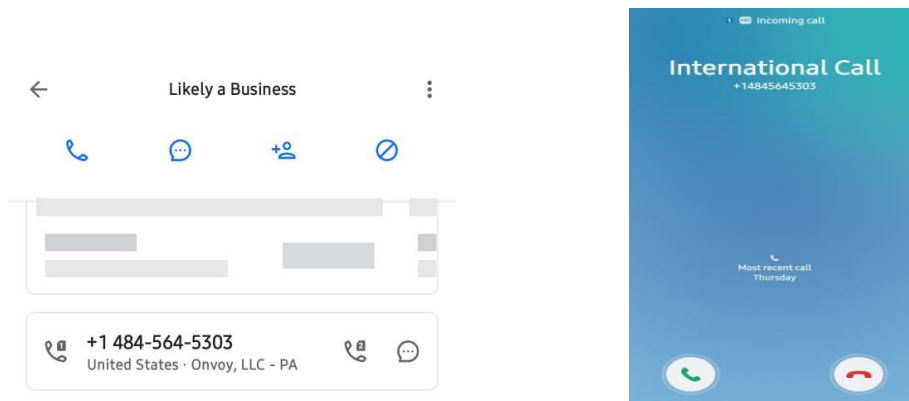
- Onboarding Assistant
- ❖ A Discord channel created for an employee



- ❖ Welcome Email is sent to the employee after onboarding



❖ The Onboarding Call is Initiated to the employee



□ Section 7: Conclusion

The implemented AI-powered HR or IS support system:

- Saves **hundreds of hours** monthly across HR/IS + employees
- Reduces cost by over **90%**
- Increases response transparency, accuracy, and SLA compliance
- Boosts **employee satisfaction**, making HR support modern, responsive, and scalable

This is a **future-ready system** that aligns with digital HR transformation goals for any modern organization.
